

Pfizer Digital & Technology Hackathon 2026

Preliminary Round Submission Prep

SafePro | Safe-by-Design AI Screening and Governance for Therapeutic Protein Discovery

Format choice: This is an HTML document because it is a polished business-style prep document featuring structured sections and slide planning.

1. Executive Summary

SafePro is a proposed safe-by-design screening and governance layer for therapeutic protein discovery. As generative algorithms accelerate pre-clinical discovery, SafePro aims to act as a pre-synthesis gatekeeper, evaluating candidate proteins to rank them by proposed developability and manufacturability while flagging potential biosecurity and toxicity risks before physical lab work begins.

Why this concept is strong for Pfizer:

- Supports responsible AI adoption in pre-clinical discovery, aligning with core values of integrity and safety.
- Aims to connect discovery and manufacturing considerations earlier in the workflow.
- Proposes to reduce resources spent on synthesizing candidates with poor predicted manufacturability.
- Acts as a triage layer that flags potential dual-use risks, incorporating human-in-the-loop review for governance.

2. Quick Facts From the Official Materials

Requirement	Details from Sources
Deliverable	5-minute video pitch. A finalized demo or working product is <i>not required</i> .
Deadline	Friday, June 19, 11:59 PM ET.
Submission Method	Unlisted YouTube link emailed to DigitalHackathonTeam@pfizer.com.

Subject Line	{Team Name} – Digital Hackathon Pitch Submission
Presentation Style	Slides and visuals are encouraged, but clarity matters most.
Pitch Coverage	Proposed solution, roadmap to MVP, alignment with Pfizer's interests/organization, key roadblocks, and expected outcome.
Evaluation Criteria	Innovation, Pfizer fit, technical feasibility, non-technical feasibility, presentation/professionalism.
IP Notice	Submitted ideas become Pfizer IP in this hackathon context.

3. Recommended Positioning

Primary problem statement: Computational pre-clinical drug development.

Secondary business value: Pharmaceutical manufacturing accuracy and/or efficiency.

Narrow scope recommendation: Biologics / therapeutic proteins, utilizing proteomics + structural biology, focused on the pre-synthesis screening step.

Rationale: The [Technical Resources](#) document explicitly asks teams to narrow their scope by pharmaceutical category, focus area, and lab workflow. Furthermore, the [Pfizer Overview](#) highlights key R&D focus areas such as vaccines, oncology, inflammation & immunology, and internal medicine. By positioning SafePro as a computational triage step prior to synthesis, the concept directly addresses the intersection of pre-clinical discovery and manufacturing efficiency.

Best Framing

One-sentence pitch:

SafePro proposes an AI screening and governance layer designed to rank AI-generated therapeutic protein candidates by developability and flag potential biosecurity risks before physical synthesis.

30-second pitch:

As generative models accelerate pre-clinical discovery, the challenge shifts toward responsible selection. SafePro is a proposed triage layer that evaluates AI-designed proteins for potential manufacturability and toxicity concerns. By aiming to provide an auditable, human-in-the-loop workflow, it seeks to help Pfizer advance the safest, most developable candidates into wet-lab workflows while supporting responsible AI adoption.

Avoid This Framing

- Do not describe the idea as a tool to "optimize virulence or toxicity."
- Do not state that the platform is built to design or generate harmful proteins.

- Do not position the concept as an active biological hazard simulator.
- Do not imply the tool allows users to manually "dial up" toxicity levels for testing.

4. The Concept

Problem

Generative models can rapidly propose novel protein structures, but these candidates may face manufacturing bottlenecks or present unintended biosecurity risks that are difficult to identify early in the process.

Solution

A proposed screening and governance triage layer that evaluates generated candidates for predicted developability and potential dual-use risks before any physical synthesis occurs.

Users

Pre-clinical computational biologists and discovery teams operating within Pfizer R&D environments.

Workflow Fit

Positioned immediately post-generation (after AI design) and pre-synthesis (before lab validation and ordering).

What Makes It Distinctive

Rather than focusing purely on finding the most active molecule, SafePro emphasizes responsible triage, aiming to ensure candidates are feasible to manufacture and safe to synthesize.

5. MVP Definition

Inputs

Protein sequence or structural data generated from upstream AI design models.

Core modules

- 1. Developability scoring: Proposes to evaluate characteristics like aggregation potential and formulation stability.
- 2. Biosecurity screening: Proposes to flag potential toxicity or hazard patterns using structural and functional data.

Decision outputs

The system aims to categorize candidates into three tiers: **Pass** (ready for synthesis), **Review** (requires human expert escalation), and **Block** (flagged for severe liabilities).

Governance layer

Aims to generate secure audit logs for all flagged assets, capturing human reviewer annotations to support responsible AI compliance.

Example workflow

A researcher submits a batch of generated biologic candidates. The MVP processes the batch, proposes developability scores, and highlights a subset for human expert review due to flagged structural patterns, allowing the team to proceed only with the most viable candidates.

6. 5-Minute Video Plan

Time	Purpose	Key Message	What to Show
0:00 - 0:45	The Hook	Highlight the challenge of triage and responsible selection in the era of rapid AI discovery.	Visual contrasting the high volume of AI designs with limited lab synthesis bandwidth.
0:45 - 1:45	The Solution	Deliver the core pitch and value proposition of SafePro as a screening layer.	Slide showing SafePro acting as a filter before lab synthesis.
1:45 - 2:45	MVP Roadmap	Outline the inputs, scoring modules, and decision outputs.	High-level block diagram of the Pass/Review/Block workflow.
2:45 - 3:45	Pfizer Alignment	Connect to relevant focus areas (e.g., vaccines/oncology) and manufacturing efficiency.	Visual linking early discovery screening directly to downstream manufacturing goals.

3:45 - 4:30	Challenges & Mitigations	Address data needs and feasibility roadblocks transparently.	Bulleted list of expected roadblocks alongside proposed mitigation strategies.
4:30 - 5:00	Conclusion	Wrap up with a strong statement on supporting safe, scalable innovation.	Team name, project title, and a final summary statement.

7. Slide-by-Slide Deck Plan

SLIDE 1

Title Slide

Goal: Introduce the team and project clearly.

- Project: SafePro
- Safe-by-Design AI Screening for Therapeutic Protein Discovery
- Team Name & Members

Suggested Visual: Clean title text aligned with a professional corporate aesthetic.

Speaker Note: Welcome the judges and state the project name and purpose.

SLIDE 2

The Pre-Clinical Challenge

Goal: Define the core problem in computational drug development.

- Rapid AI generation outpaces lab synthesis bandwidth
- Manufacturing bottlenecks emerge late in development
- Safety and governance review can lag behind rapid candidate generation

Suggested Visual: Diagram showing a bottleneck between "AI Generation" and "Lab Synthesis."

Speaker Note: Emphasize that generating candidates is no longer the main hurdle; responsibly selecting them is.

SLIDE 3

The SafePro Solution

Goal: Explain the concept and value proposition.

- Pre-synthesis triage layer
- Aims to rank candidates by developability and manufacturability
- Proposes to flag potential biosecurity and toxicity risks

Suggested Visual: A filtering funnel graphic labeled "SafePro" separating designs from the lab.

Speaker Note: Introduce SafePro as a governance tool that connects discovery with manufacturing considerations.

SLIDE 4

MVP Workflow & Roadmap

Goal: Demonstrate technical feasibility and architecture.

- Inputs: Protein sequences and structural data
- Scoring modules: Developability and Biosecurity evaluation
- Outputs: Pass, Review (Human-in-the-loop), Block

Suggested Visual: Flowchart illustrating the decision logic and audit trail generation.

Speaker Note: Walk through how data moves through the MVP to create actionable triage decisions.

SLIDE 5

Alignment with Pfizer

Goal: Demonstrate clear Pfizer fit.

- Supports biologics and vaccines pre-clinical pipelines
- Aims to de-risk downstream manufacturing at Global Supply Sites
- Aligns with Pfizer's emphasis on integrity, transparency, and responsible decision-making

Suggested Visual: Icons representing R&D and supply chain integration.

Speaker Note: Detail how this supports Pfizer's specific focus areas and organizational values.

SLIDE 6

Key Roadblocks & Mitigations

Goal: Address non-technical feasibility and implementation realism.

- Challenge: Access to robust training data
- Mitigation: Leverage public protein databases initially, scaling to internal assay data

- Challenge: Managing false positives
- Mitigation: Implement human-in-the-loop expert review

Suggested Visual: Two-column layout comparing Challenges and Mitigations.

Speaker Note: Acknowledge that data acquisition is a hurdle, but outline a logical path forward.

SLIDE 7

Expected Outcomes

Goal: Leave a lasting impression of the project's potential impact.

- Aim to reduce late-stage attrition
- Foster safe pre-clinical discovery
- Support responsible AI scaling

Suggested Visual: SafePro concept title alongside a final summary badge.

Speaker Note: Conclude by summarizing that SafePro is about safe, scalable innovation.

8. Judging Criteria

Criterion	How SafePro Aligns
Innovation	Proposes a governance and safety screening approach rather than merely focusing on molecule generation.
Pfizer Fit	Directly targets computational pre-clinical drug development and aims to address manufacturing bottlenecks early.
Technical Feasibility	Aims to leverage existing structural and biological data paradigms to build a modular scoring framework.
Non-Technical Feasibility	Integrates human review and audit logs to proactively address governance, regulatory, and safety concerns.
Presentation	Designed to be clear, structured, and strictly adherent to the 5-minute pitch requirement without claiming false metrics.

9. Likely Judge Questions and Strong Answers

- What data would you use to train this?

Answer: We would aim to utilize public protein databases and literature annotations initially,

exploring the integration of internal proprietary assay data in a later pilot phase.

- **How does this align with Pfizer's pipeline?**

Answer: It targets Stage 1 discovery, aiming to de-risk candidate molecules before they advance toward clinical trials and manufacturing.

- **Is a working demo required today?**

Answer: No, the submission guidelines explicitly state a finalized demo is not required for this preliminary round.

- **Why include a human-in-the-loop?**

Answer: To intelligently manage potential false positives and ensure expert oversight on any candidates flagged for biosecurity risks.

- **How does this address manufacturing?**

Answer: By aiming to filter out candidates with poor predicted developability characteristics before physical lab resources are expended.

- **Could this tool be misused?**

Answer: We propose strict access controls, secure audit logs, and a purely defensive screening posture to mitigate potential misuse.

- **Why combine developability scoring with governance in one platform?**

Answer: Because Pfizer is not only selecting promising candidates; it is also managing how those candidates move into real workflows. Combining both functions keeps scientific triage and responsible review connected in one decision process.

- **What is the main roadblock to implementation?**

Answer: Securing high-quality, normalized training data for developability scoring is challenging, which is why we propose a phased module rollout.

10. Submission Checklist

Content Prep

- ☐ Deck finalized (exactly 7 slides).
- ☐ Pitch script prepared and rehearsed.

Recording

- ☐ Audio and video quality tested.
- ☐ Final video length verified (must be under 5 minutes).

YouTube Upload

- ☐ Video uploaded to YouTube.
- ☐ Visibility setting confirmed as **Unlisted**.
- ☐ Link tested in an Incognito/Private window.

Email Submission

- ☐ Addressed to DigitalHackathonTeam@pfizer.com.
- ☐ Subject line formatted correctly: {Team Name} – Digital Hackathon Pitch Submission.

11. Ready-to-Send Email Template

To: DigitalHackathonTeam@pfizer.com

Subject: [Insert Team Name] - Digital Hackathon Pitch Submission

Hello Hackathon Team,

Please find our preliminary round submission for the Pfizer Digital & Technology Hackathon below.

Team Name: [Insert Team Name]

Project Title: SafePro - Safe-by-Design AI Screening for Therapeutic Protein Discovery

YouTube Unlisted Link: [Insert Link Here]

Our pitch addresses the "Computational pre-clinical drug development" problem statement. We propose a safe-by-design screening and governance layer that evaluates AI-generated therapeutic protein candidates for developability, manufacturability, and potential biosecurity concerns before synthesis.

Thank you for your time and consideration.

Best regards,

[Your Name / Team Name]

12. Final Coaching Notes

Emphasize

Highlight the goal of responsible AI adoption, pre-clinical triage, and strong alignment with Pfizer's emphasis on integrity, transparency, and operational fit. Focus on the proposed connection between early computational discovery and downstream manufacturing feasibility.

Avoid Saying

Do not claim guaranteed cost savings, do not use fabricated accuracy statistics, and do not position the tool as a method for actively designing hazards.

Delivery Tips

Speak clearly and pace yourself to comfortably fit within the 5-minute limit. As the guidelines state, clarity of thought matters most—focus on delivering your message rather than worrying

about high-end video production.

13. Sources

- [Submission Guidelines \(@u.northwestern.edu Mail - Pfizer Digital ...s.pdf\)](#)
- [Technical Resources - Pfizer Digital & Technology Hackathon 2026.pdf](#)
- [Pfizer Overview - Pfizer Digital & Technology Hackathon 2026.pdf](#)
- [Problem Statements - Pfizer Digital & Technology Hackathon 2026.pdf](#)